

This strategy function combines the RSI calculation with trading logic to generate position signals. By encapsulating each component in its own function, we've created a modular, readable system that can be easily tested and refined.

Creating a Function Library: Your Trading Toolkit

As you develop more functions, you'll want to organize them into a reusable library. This approach is used by all professional trading firms to maintain and leverage their intellectual property.

Here's how you might organize a basic trading function library:

```
1. # indicators.py - Technical indicator functions
2.
3. def calculate_sma(prices, period):
4.     """Calculate Simple Moving Average"""
5.     return sum(prices[-period:]) / period
6.
7. def calculate_ema(prices, period, smoothing=2):
8.     """Calculate Exponential Moving Average"""
9.     ema = [sum(prices[:period]) / period] # Initialize with SMA
10.
11.     multiplier = smoothing / (period + 1)
12.
13.     for price in prices[period:]:
14.         ema.append((price - ema[-1]) * multiplier + ema[-1])
15.
16.     return ema[-1]
17.
18. def calculate_rsi(prices, period=14):
19.     """Calculate Relative Strength Index"""
20.     # Implementation as shown earlier
21.     pass
22.
23. # signals.py - Trading signal functions
24.
25. def check_golden_cross(prices, fast_period=50, slow_period=200):
```